1. Question

Your organization needs to implement a scalable and efficient solution for handling IP addressing and subnets for its expanding network infrastructure. Which advanced IP Services technology best addresses the need to simplify address management and improve address utilization?
Static IP Allocation
IP Address Management (IPAM)
Dynamic Host Configuration Protocol (DHCP)
Network Address Translation (NAT)

The correct answer is:

✅ **IP Address Management (IPAM)**

Why?

IP Address Management helps organizations centrally manage, track, plan, and organize IP addresses across large networks.

It improves:

- **Address utilization** (avoids wasted IPs)
- **Subnet management**
- **Scalability**
- **Automation**
- **Conflict detection**

Why not the others?

- **Static IP Allocation** → Manual and hard to scale
- **DHCP** → Automatically assigns IPs, but does not provide full address planning/management
- **NAT** → Conserves public IPs, but not for subnet/IP management

✅ **Final Answer: IP Address Management (IPAM)**

2. **2. Question**

Which option is a multicast MAC address in IPv6 used for neighbor discovery protocols?
01:00:5e:00:00:fb
33:33:ff:01:02:03
ff:ff:ff:ff:ff:ff
08:00:27:14:73:0a

The correct answer is:

✅ **33:33:ff:01:02:03**

Why?

In IPv6, multicast MAC addresses used for Neighbor Discovery Protocol (NDP) begin with:

```
33:33
```

The address:

```
33:33:ff:01:02:03
```

is an IPv6 multicast MAC address commonly used for:

- Neighbor Solicitation
- Neighbor Advertisement
- Other IPv6 multicast communications

Why not the others?

- `01:00:5e:00:00:fb` → IPv4 multicast MAC address
- `ff:ff:ff:ff:ff:ff` → Broadcast MAC address
- `08:00:27:14:73:0a` → Normal unicast MAC address

✅ **Final Answer: 33:33:ff:01:02:03**

3. Question

On a Cisco router, which configuration command can be used to enable the IP SLA responder but only for certain types of IP SLAs, such as UDP Echo, and not for others?
ip sla responder tcp-connect only
ip sla responder udp-echo
ip sla responder
ip sla responder udp-echo only

The correct answer is:

✅ **ip sla responder udp-echo**

Why?

On a Cisco router, the command:

```
ip sla responder udp-echo
```

enables the router to respond only to **UDP Echo** IP SLA operations instead of enabling all responder functions.

Why not the others?

- `ip sla responder` → Enables all responder functions
- `ip sla responder tcp-connect only` → Invalid command
- `ip sla responder udp-echo only` → Incorrect syntax

✅ **Final Answer: ip sla responder udp-echo**

4. Question

You are configuring a multilayer switch to optimize the network access policies for a critical segment in your enterprise. Which of the following methods would allow you to provide advanced security features at Layer 2 and Layer 3, such as dynamic VLAN assignments based on user credentials and control over traffic with Access Control Lists (ACLs), while also ensuring that non-compliant devices are quarantined to a remediation VLAN?
Utilizing Dynamic Trunking Protocol (DTP) with port security
Setting up Spanning Tree Protocol (STP) with root guard and portfast
Enabling DHCP snooping and ARP inspection
Implementing 802.1X authentication with VLAN ID assignment and downloadable ACLs

The correct answer is:

✅ **Implementing 802.1X authentication with VLAN ID assignment and downloadable ACLs**

Why?

IEEE 802.1X provides:

- **User/device authentication**
- **Dynamic VLAN assignment**
- **Downloadable ACLs (dACLs)**
- **Quarantine/remediation VLANs** for non-compliant devices
- Layer 2 and Layer 3 access control

This is commonly integrated with systems like:

- RADIUS servers
- Cisco ISE
- NAC (Network Access Control)

Why not the others?

- **DTP with port security** → Helps with switchport security, not dynamic user-based policies

- **STP with root guard/portfast** → Prevents Layer 2 loops, not authentication or remediation
- **DHCP snooping and ARP inspection** → Protects against spoofing attacks but does not provide user-based VLAN assignment or downloadable ACLs

✅ **Final Answer: Implementing 802.1X authentication with VLAN ID assignment and downloadable ACLs**

5. Question

In IEEE 802.1Q VLAN tagging, how is the VLAN information carried over the Ethernet frame?
It replaces the original Ethernet frame header.
It is appended at the end of the Ethernet frame without any modifications.
It overwrites the data payload in the Ethernet frame.
It is inserted between the source MAC address and Ethertype fields.

The correct answer is:

✅ **It is inserted between the source MAC address and Ethertype fields.**

Why?

IEEE 802.1Q adds a **4-byte VLAN tag** inside the Ethernet frame.

The tag is inserted:

- After the **Source MAC Address**
- Before the **Ethertype field**

Ethernet Frame Structure with 802.1Q

```
Destination MAC | Source MAC | 802.1Q Tag | Ethertype | Data | FCS
```

Why not the others?

- It does **not replace** the Ethernet header
- It is **not appended** to the end
- It does **not overwrite** the payload

✅ **Final Answer: It is inserted between the source MAC address and Ethertype fields.**

6. Question

You have configured PPPoE on a network. Which two of the following protocols would you expect to be used during the process to establish a connection and authenticate the user?
NCP and PAP

LCP and CHAP
ARP and PAP
DNS and CHAP

The correct answers are:

✅ **LCP and CHAP**

Why?

Point-to-Point Protocol over Ethernet (PPPoE) uses PPP protocols during connection establishment.

Two important protocols are:

1. **LCP (Link Control Protocol)**
   a. Establishes, configures, and tests the PPP connection
2. **CHAP (Challenge Handshake Authentication Protocol)**
   a. Authenticates the user securely

Why not the others?

- **NCP and PAP** → NCP configures network layer protocols after authentication
- **ARP and PAP** → ARP is unrelated to PPP authentication
- **DNS and CHAP** → DNS resolves names, not PPP session setup

✅ **Final Answer: LCP and CHAP**

7. Question

Which command can be used to verify conditional forwarding, understanding how traffic is being forwarded based on policy-based routing (PBR) within a Cisco router's configuration?
show ip route
show policy-map
show access-lists
show ip policy

✅ **show ip policy**

Why?

On a Cisco router, the command:

```
show ip policy
```

is used to verify:

- Which interfaces have Policy-Based Routing (PBR) applied
- How traffic is being conditionally forwarded based on policies

It helps confirm that routing decisions are being made according to configured route maps and policies instead of normal routing tables.

Why not the others?

- `show ip route` → Shows routing table only
- `show policy-map` → Used mainly for QoS policies
- `show access-lists` → Displays ACL entries, not PBR application status

✅ **Final Answer: show ip policy**

8. Question

During the configuration of a switchport, which port security mode will not disable the port but will log the security violation and drop the offending packet?
Shutdown
Protect
Monitor

The correct answer is:

✅ **Protect**

Why?

On a Cisco switch, the **Protect** port-security violation mode:

- Drops packets from unauthorized MAC addresses
- Does **not** shut down the port
- Keeps the port operational

Port Security Modes

- **Shutdown** → Disables the port (err-disabled state)
- **Protect** → Drops offending traffic silently without shutting down
- **Restrict** (not listed) → Drops packets and logs/counts violations
- **Monitor** → Not a valid Cisco port-security violation mode

⚠️ Note: The question says "log the security violation and drop the offending packet," which technically matches **Restrict** mode. However, since it is not one of the provided options, the best available answer is:

✅ **Protect**

9. Question

You are configuring DHCP snooping on a switch. Which of the following commands would allow you to specify a particular VLAN for DHCP snooping?
ip dhcp snooping enable
ip dhcp snooping vlan
switchport dhcp snooping vlan
ip dhcp guard enable vlan

✅ **ip dhcp snooping vlan**

Why?

On a Cisco switch, the command:

```
ip dhcp snooping vlan <vlan-id>
```

enables DHCP snooping for a specific VLAN.

Example:

```
ip dhcp snooping vlan 10
```

Why not the others?

- `ip dhcp snooping enable` → Incorrect Cisco syntax
- `switchport dhcp snooping vlan` → Invalid command
- `ip dhcp guard enable vlan` → Not used for DHCP snooping configuration

✅ **Final Answer: ip dhcp snooping vlan**

10. Question

Which of the following security protocols uses asymmetric encryption to provide secure key exchange in the encapsulation of VPN data?
IPsec
SSH
IKEv2
SSL/TLS

The correct answer is:

✅ **IKEv2**

Why?

Internet Key Exchange version 2 (IKEv2) is responsible for:

- Securely negotiating VPN security parameters
- Performing **asymmetric encryption–based key exchange**
- Establishing shared session keys for IPsec VPN tunnels

It commonly uses:

- RSA certificates
- Diffie-Hellman key exchange

Why not the others?

- **IPsec** → Encrypts VPN traffic but relies on IKE for key exchange
- **SSH** → Secure remote access protocol, not VPN encapsulation
- **SSL/TLS** → Uses asymmetric encryption too, but not specifically for IPsec VPN key exchange in this context

✅ **Final Answer: IKEv2**

11. Question

In an IPv6-enabled network, an engineer has configured an OSPFv3 router with the following commands: interface GigabitEthernet0/0 ipv6 ospf 1 area 0 exit ipv6 router ospf 1 router-id 1.1.1.1 Despite this configuration, OSPF neighbors are not being formed. Which of the following could be the most likely reason?
The router-id must be configured under the interface, not under the OSPF process.
IPv6 routing is not enabled globally on the router.
The interface GigabitEthernet0/0 must have a manually assigned IPv6 address.
The OSPFv3 process number must match the interface number.

The correct answer is:

✅ **IPv6 routing is not enabled globally on the router.**

Why?

For OSPFv3 to work on a Cisco router, IPv6 routing must be enabled globally with:

```
ipv6 unicast-routing
```

Without this command, the router may have IPv6 interface commands configured, but it will not properly route IPv6 traffic or form OSPFv3 adjacencies.

Why not the others?

- Router ID is correctly configured under `ipv6 router ospf 1`
- OSPFv3 can use link-local IPv6 addresses, so a manual global IPv6 address is not always required
- OSPF process number does not need to match the interface number

✅ **Final Answer: IPv6 routing is not enabled globally on the router.**

12. Question

In a Cisco network, which BGP attribute influences the inbound traffic path from an outside AS (Autonomous System)?
AS Path
Local Preference
MED (Multi-Exit Discriminator)
Weight

The correct answer is:

✅ **MED (Multi-Exit Discriminator)**

Why?

Border Gateway Protocol uses the **MED** attribute to suggest to an external AS which path should be preferred for inbound traffic into the AS.

- Lower MED value = more preferred path
- Used between neighboring autonomous systems

Why not the others?

- **AS Path** → Primarily prevents loops and influences outbound path selection
- **Local Preference** → Influences outbound traffic within an AS
- **Weight** → Cisco proprietary attribute affecting outbound traffic locally only

✅ **Final Answer: MED (Multi-Exit Discriminator)**

13. Question

In an IPv6 network, what is the purpose of the Next Header field in the IPv6 header, and which value of this field indicates that a Hop-by-Hop Options header is present?
The Next Header field specifies the length of the payload. A value of 58 indicates a Hop-by-Hop Options header.

The Next Header field denotes the number of hops remaining. A value of 2 indicates a Hop-by-Hop Options header.
The Next Header field specifies the type of header subsequent to the IPv6 header. A value of 44 indicates a Hop-by-Hop Options header.
The Next Header field specifies the type of header immediately following the IPv6 header. A value of 0 indicates a Hop-by-Hop Options header.

The correct answer is:

✅ **The Next Header field specifies the type of header immediately following the IPv6 header. A value of 0 indicates a Hop-by-Hop Options header.**

Why?

In Internet Protocol version 6, the **Next Header** field:

- Identifies the header that comes next after the IPv6 header
- Can point to:
    - TCP
    - UDP
    - ICMPv6
    - Extension headers

A value of:

```
0
```

indicates a **Hop-by-Hop Options** extension header.

Why not the others?

- 58 = ICMPv6
- 44 = Fragment Header
- The field does not specify payload length or hop count

✅ **Final Answer: The Next Header field specifies the type of header immediately following the IPv6 header. A value of 0 indicates a Hop-by-Hop Options header.**

14. Question

In the context of network security, what is the primary purpose of implementing a Demilitarized Zone (DMZ) on a corporate network?
To provide a secure network segment for internal users to access external resources directly without any monitoring.
To isolate and protect the internal network from untrusted external networks while still allowing limited services to be accessed from the outside.

To encrypt data traffic between internal and external users, ensuring confidentiality and integrity of the data.
To ensure high availability of internal resources by distributing the traffic load among multiple servers in the internal network.

The correct answer is:

✅ **To isolate and protect the internal network from untrusted external networks while still allowing limited services to be accessed from the outside.**

Why?

A Demilitarized zone (DMZ) is a separate network segment placed between:

- The internal trusted network
- The external untrusted network (Internet)

It is used to host public-facing services such as:

- Web servers
- Email servers
- DNS servers

This design helps protect the internal network even if a public-facing server is compromised.

Why not the others?

- DMZs do not allow unrestricted direct access without monitoring
- Encryption is handled by protocols like SSL/TLS or IPsec
- Load balancing/high availability is a different function

✅ **Final Answer: To isolate and protect the internal network from untrusted external networks while still allowing limited services to be accessed from the outside.**

15. Question

Which security feature on Cisco IOS software provides protection from ARP spoofing or poisoning attacks by associating a static ARP entry with a specific interface?
EtherChannel Guard
Dynamic ARP Inspection (DAI)
Port Security
ARP Inspection

The correct answer is:

✅ **Dynamic ARP Inspection (DAI)**

Why?

Cisco **Dynamic ARP Inspection (DAI)** helps protect against:

- ARP spoofing
- ARP poisoning attacks

It validates ARP packets against trusted information such as the DHCP snooping binding table and can prevent malicious ARP responses from being forwarded.

Why not the others?

- **EtherChannel Guard** → Protects EtherChannel configurations
- **Port Security** → Controls MAC addresses on switch ports
- **ARP Inspection** → Not the official Cisco feature name

✅ **Final Answer: Dynamic ARP Inspection (DAI)**

I prefer this response

16. Question

Which of the following protocols offers both authentication and encryption, ensuring secure data transmission over IP networks by using a mechanism that operates at the Network Layer?
IPSec
SSL/TLS
SSH
802.1X

The correct answer is:

✅ **IPSec**

Why?

Internet Protocol Security (IPSec) operates at the **Network Layer (Layer 3)** and provides:

- Authentication
- Encryption
- Data integrity
- Secure IP communication

It is commonly used for:

- VPNs
- Site-to-site tunnels
- Secure remote access

Why not the others?

- **SSL/TLS** → Operates mainly at the Transport/Application layers
- **SSH** → Secure remote login protocol
- **802.1X** → Network access control/authentication protocol

✅ **Final Answer: IPSec**

17. Question

Which of the following statements about the OSI model is true regarding the function of the Network layer?
It handles the end-to-end delivery of data, including error checking and flow control.
It encrypts data to ensure secure communication between devices.
It establishes, manages, and terminates sessions between two communicating hosts.
It determines the path through which data should be transmitted from source to destination

The correct answer is:

✅ **It determines the path through which data should be transmitted from source to destination**

Why?

In the OSI model, the **Network Layer (Layer 3)** is responsible for:

- Routing
- Path determination
- Logical addressing (IP addresses)
- Packet forwarding between networks

Routers operate primarily at this layer.

Why not the others?

- End-to-end delivery, flow control, and error checking → Transport Layer
- Encryption → Usually Presentation Layer
- Session establishment/management → Session Layer

✅ **Final Answer: It determines the path through which data should be transmitted from source to destination**

18. Question

In a multi-VLAN environment, a network administrator needs to ensure VLAN 10 and VLAN 20 can communicate between each other while also isolating other VLANs. Which of the following configurations would best achieve this requirement?
Implement Routed Access Layer using Layer 3 switches.
Configure ACLs to permit traffic between VLAN 10 and VLAN 20 on Layer 2 switches.
Use Private VLANs (PVLAN) to isolate traffic between VLANs.
Configure a router with sub-interfaces for each VLAN.

The correct answer is:

✅ **Implement Routed Access Layer using Layer 3 switches.**

Why?

Using a Layer 3 switching routed access layer allows:

- Inter-VLAN routing between VLAN 10 and VLAN 20
- Traffic control and isolation for other VLANs
- Better scalability and performance in enterprise networks

A Layer 3 switch can:

- Route between selected VLANs
- Apply ACLs to control communication
- Keep unwanted VLANs isolated

Why not the others?

- ACLs on Layer 2 switches → Layer 2 switches normally cannot route between VLANs
- Private VLANs → Mainly isolate devices within the same VLAN
- Router-on-a-stick (sub-interfaces) → Works, but less scalable and efficient than Layer 3 switching in enterprise environments

✅ **Final Answer: Implement Routed Access Layer using Layer 3 switches.**

19. Question

What is the purpose of the PVID (Port VLAN ID) in a switch configuration, and how does it influence network access?
The PVID indicates the priority of the port in spanning tree calculations.
The PVID is used to tag frames received on the port with the specified VLAN ID if they are untagged.
The PVID is used to assign a unique identifier to the switch port for security purposes.
The PVID prevents frames with the specified VLAN ID from being transmitted on the port.

The correct answer is:

✅ **The PVID is used to tag frames received on the port with the specified VLAN ID if they are untagged.**

Why?

Port VLAN ID (PVID) defines the default VLAN assigned to incoming untagged frames on a switch port.

When an untagged frame arrives:

- The switch assigns it to the VLAN specified by the PVID
- This determines which VLAN/network the device can access

Why not the others?

- Spanning Tree priority is unrelated to PVID
- PVID is not a security identifier
- PVID does not block VLAN traffic

✅ **Final Answer: The PVID is used to tag frames received on the port with the specified VLAN ID if they are untagged.**

20. Question

When configuring an IPv6 address on a router interface, which of the following command sequences will enable IPv6 and assign a global address to the interface?
interfaceipv6 address 2001:db8::1/64 ipv6 enable
ipv6 unicast-routing interfaceipv6 address 2001:db8::1/64
ipv6 enable ipv6 address 2001:db8::1/64
ipv6 address 2001:db8::1/64 ipv6 enable

The correct answer is:

✅

```
interface <interface-name>
 ipv6 address 2001:db8::1/64
 ipv6 enable
```

Why?

On a Cisco router:

- `ipv6 address 2001:db8::1/64` assigns the global IPv6 address
- `ipv6 enable` enables IPv6 processing on the interface and creates a link-local address if needed

Typical configuration:

```
interface GigabitEthernet0/0
 ipv6 address 2001:db8::1/64
 ipv6 enable
```

Important Note

To route IPv6 traffic globally, you also usually need:

```
ipv6 unicast-routing
```

Why not the others?

- `ipv6 unicast-routing interface` → Invalid syntax
- Missing `interface` context → Incorrect configuration structure
- Commands outside interface configuration mode → Invalid

✅ **Final Answer:**

```
interface <interface-name>
 ipv6 address 2001:db8::1/64
 ipv6 enable
```

21. Question

In an IPv6 network, what is the purpose of the flow label field in the IPv6 header, and how is it utilized?
It is reserved for future use and currently not utilized in IPv6 networks.
It is primarily used for security purposes to label encrypted traffic.
It is used to differentiate packets belonging to the same flow to enable routers to prioritize their sequencing.
It is used to mark packets for Quality of Service (QoS) for applications like VoIP

The correct answer is:

✅ **It is used to differentiate packets belonging to the same flow to enable routers to prioritize their sequencing.**

Why?

In Internet Protocol version 6, the **Flow Label** field:

- Identifies packets belonging to the same traffic flow
- Helps routers handle packets consistently
- Supports efficient processing for real-time traffic such as:

  - VoIP
  - Streaming
  - Video conferencing

Routers can use this field to recognize and process packets belonging to the same communication session.

Why not the others?

- It is not simply reserved/unused
- It is not primarily for encryption/security
- QoS can benefit from the flow label, but its main purpose is identifying packets within the same flow

✅ **Final Answer: It is used to differentiate packets belonging to the same flow to enable routers to prioritize their sequencing.**

22. Question

Which 802.1X authentication method allows for the greatest flexibility in supporting various authentication mechanisms, such as passwords, digital certificates, and smart cards?
EAP-TLS
PEAP
LEAP
EAP-MD5

The correct answer is:

✅ **PEAP**

Why?

Protected Extensible Authentication Protocol (PEAP) provides the greatest flexibility because it:

- Creates a secure TLS tunnel
- Supports multiple inner authentication methods
- Can work with:
  - Passwords
  - Certificates
  - Smart cards
  - Token-based authentication

It is widely used in enterprise 802.1X deployments.

Why not the others?

- **EAP-TLS** → Very secure but mainly certificate-based
- **LEAP** → Older Cisco protocol with known security weaknesses
- **EAP-MD5** → Only basic password authentication and no strong protection

✅ **Final Answer: PEAP**

23. Question

Which of the following commands will display the state of an interface's Line Protocol and IP address configuration on a Cisco router?
show running-config interface
show protocols
show ip interface brief
show interface status
The correct answer is:

✅ **show ip interface brief**

Why?

On a Cisco router, the command:

```
show ip interface brief
```

displays:

- Interface status
- Line protocol status
- Assigned IP addresses
- Quick operational summary of interfaces

Example output includes:

```
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up
up
```

Why not the others?

- `show running-config interface` → Shows configuration only
- `show protocols` → Shows protocol information but less concise for IP/interface overview
- `show interface status` → Commonly used on switches, not routers

✅ **Final Answer: show ip interface brief**

24. Question

In the context of 802.1X port-based authentication for a network access setup, which component is responsible for providing the authentication credentials and managing network policies for the client device?
Authenticator
Network Access Control
Supplicant
Authentication Server
The correct answer is:

✅ **Supplicant**

Why?

In IEEE 802.1X authentication:

- **Supplicant** → The client device/software that provides authentication credentials
- **Authenticator** → Usually the switch or wireless access point controlling access
- **Authentication Server** → Typically a RADIUS server validating credentials

The supplicant is responsible for:

- Sending usernames/passwords or certificates
- Communicating with the authenticator
- Participating in the authentication process

Why not the others?

- **Authenticator** → Controls access, not client credentials
- **Network Access Control** → General security concept, not the specific client role
- **Authentication Server** → Verifies credentials and applies policies

✅ **Final Answer: Supplicant**

25. Question

An engineer is troubleshooting a network and notices that some of the routers are not forming OSPF neighbor relationships. The network consists of both broadcast and non-broadcast multi-access (NBMA) segments. Which of the following statements is a likely reason for the issue?
The OSPF router IDs are not unique in the network.
The OSPF priority values are set to zero on all routers.
The OSPF Hello and Dead timers are not matching between routers.
The OSPF network types are not properly configured or matching on the NBMA network.

The correct answer is:

✅ **The OSPF network types are not properly configured or matching on the NBMA network.**

Why?

Open Shortest Path First neighbor relationships on NBMA networks are sensitive to:

- Network type configuration
- Neighbor statements
- Multicast behavior

If routers on an NBMA segment have mismatched OSPF network types (broadcast, non-broadcast, point-to-multipoint, etc.), they may fail to form adjacencies.

Why not the others?

- Duplicate router IDs can cause issues, but NBMA specifically points toward network type configuration
- Priority 0 only prevents DR/BDR election participation
- Hello/Dead timer mismatches can break adjacencies, but NBMA network-type mismatch is the more targeted issue in this scenario

✅ **Final Answer: The OSPF network types are not properly configured or matching on the NBMA network.**

26. Question

In a Cisco network, which DHCP option can be used to specify a bootfile and TFTP server to a client that requires special configurations, such as a PXE boot?
Option 67
Option 43
Option 66
Option 150
The correct answer is:

✅ **Option 67**

Why?

In Cisco DHCP configurations:

- **Option 66** → Specifies the TFTP server name/address
- **Option 67** → Specifies the bootfile name

For PXE boot environments, the bootfile itself is provided using:

```
Option 67
```

Example:

`pxelinux.0`

Why not the others?

- **Option 43** → Vendor-specific information
- **Option 66** → TFTP server only
- **Option 150** → Commonly used by Cisco IP phones for TFTP servers

✅ **Final Answer: Option 67**

27. Question

Which of the following security measures can provide protection against switch spoof attacks on a Cisco network?
Port Security
DHCP Snooping
Private VLANs (PVLANs)
Dynamic ARP Inspection (DAI)
The correct answer is:

✅ **Port Security**

Why?

Cisco **Port Security** helps protect against switch spoofing attacks by:

- Limiting MAC addresses allowed on a switch port
- Preventing unauthorized devices from connecting
- Restricting rogue switches from participating in the network

It is commonly used on access ports to stop unauthorized switch connections.

Why not the others?

- **DHCP Snooping** → Protects against rogue DHCP servers
- **PVLANs** → Isolate devices within VLANs
- **Dynamic ARP Inspection (DAI)** → Protects against ARP spoofing attacks

✅ **Final Answer: Port Security**

28. Question

Which of the following IPv6 address types is assigned automatically to an interface using Stateless Address Autoconfiguration (SLAAC)?
Multicast Address
Link-Local Address
Global Unicast Address
Anycast Address

The correct answer is:

✅ **Global Unicast Address**

Why?

Stateless Address Autoconfiguration (SLAAC) allows a device to automatically generate its own:

- **Global Unicast IPv6 address**
  using router advertisements from an IPv6 router.

The router provides:

- Network prefix
- Default gateway information

The host then creates its own interface identifier.

Important Note

A **Link-Local Address** is also automatically generated in IPv6, but SLAAC specifically refers to the automatic creation of routable global unicast addresses using router advertisements.

Why not the others?

- Multicast → Used for one-to-many communication
- Link-Local → Auto-generated independently of SLAAC process
- Anycast → Shared address among multiple devices

✅ **Final Answer: Global Unicast Address**

29. Question

You have configured HSRP in your network to provide redundancy for the default gateway on your LAN. Router A is configured with a priority of 110, and Router B is configured with a priority of 100. Both routers are currently active and functioning. Suddenly, Router A fails and then comes back up. What will be the state of Router B after Router A recovers?
Router B will enter the Standby state, and Router A will become the Active router.
Router B will enter the Init state while Router A becomes the Active router.
Router B will remain in the Active state, and Router A will enter the Standby state.
Both routers will remain in Active state, leading to an IP address conflict.

The correct answer is:

✅ **Router B will enter the Standby state, and Router A will become the Active router.**

Why?

Hot Standby Router Protocol (HSRP) selects the router with the **higher priority** as the Active router.

- Router A priority = 110
- Router B priority = 100

When Router A recovers, it will typically reclaim the Active role (assuming **preemption** is enabled, which is common in HSRP redundancy designs).

Result:

- Router A → Active
- Router B → Standby

Why not the others?

- Init state is only during startup/initialization
- Two Active routers would create a conflict and is not normal HSRP behavior
- Lower-priority router does not normally stay Active after higher-priority router preempts

✅ **Final Answer: Router B will enter the Standby state, and Router A will become the Active router.**

30. Question

In a given network topology with a subnetted IP addressing scheme, if a router with an IP address of 192.168.1.1/26 has an incorrect subnet mask configured as /25, which of the following problems may arise when this router attempts to route traffic?
The router will broadcast the subnet conflict and ARP requests will fail.
The router will be able to route to other subnets but will not communicate with devices on its own subnet.
The router will not be able to route traffic to any other subnet.
The router will route traffic incorrectly within its own subnet and misroute packets to the internet.
The correct answer is:

✅ **The router will route traffic incorrectly within its own subnet and misroute packets to the internet.**

Why?

A router configured with:

- Correct subnet should be:

```
192.168.1.1/26
```

- But configured as:

/25
will incorrectly believe that more hosts are part of its local subnet than actually are.

For example:

- /26 range → 192.168.1.0–192.168.1.63
- /25 range → 192.168.1.0–192.168.1.127

This causes the router to:

- Treat some remote hosts as local
- Send ARP requests instead of routing packets properly
- Misroute traffic inside the subnet structure

Why not the others?

- The router can still route traffic in many cases
- ARP itself does not completely fail
- The issue is primarily incorrect local subnet determination

✅ **Final Answer: The router will route traffic incorrectly within its own subnet and misroute packets to the internet.**

31. Question

In a network utilizing DHCP snooping, which advanced configuration ensures that only legitimate DHCP server traffic reaches the clients while malicious DHCP server attempts are blocked?
Setting up Dynamic ARP Inspection along with DHCP snooping
Enabling DHCP snooping on all network switches
Defining trusted and untrusted ports for DHCP server traffic
Configuring IP source guard on all ports in the VLAN

The correct answer is:

✅ **Defining trusted and untrusted ports for DHCP server traffic**

Why?

In Cisco DHCP snooping:

- **Trusted ports** are allowed to send DHCP server responses
- **Untrusted ports** cannot send DHCP offer/ack packets

This prevents rogue or malicious DHCP servers from distributing incorrect IP information to clients.

Typical design:

- Uplink to real DHCP server → Trusted
- Client access ports → Untrusted

Why not the others?

- Dynamic ARP Inspection works with DHCP snooping but focuses on ARP attacks
- Enabling DHCP snooping alone is not enough without trusted/untrusted configuration
- IP Source Guard protects against IP spoofing, not rogue DHCP servers directly

✅ **Final Answer: Defining trusted and untrusted ports for DHCP server traffic**

32. Question

When configuring a switch to handle large, scalable network environments, which feature allows for VLAN configurations to be propagated across multiple switches dynamically?
PortFast
VTP (VLAN Trunking Protocol)
Trunking
STP (Spanning Tree Protocol)
The correct answer is:

✅ **VTP (VLAN Trunking Protocol)**

Why?

VLAN Trunking Protocol allows VLAN information to be:

- Dynamically propagated
- Synchronized across multiple switches
- Centrally managed in large switched networks

This reduces manual VLAN configuration on every switch.

Why not the others?

- **PortFast** → Speeds up access port transition to forwarding state
- **Trunking** → Carries multiple VLANs across links but does not distribute VLAN databases
- **STP** → Prevents Layer 2 loops

✅ **Final Answer: VTP (VLAN Trunking Protocol)**

33. Question

In the context of IPv6 addressing, what is the purpose of the Global Unicast Address (GUA), and how can you identify it from other IPv6 address types?

Its purpose is for communication within a unique local network, identified by the prefix fc00::/7.
Its purpose is for expressing multicast traffic, identified by the prefix ff00::/8.
Its purpose is for global communication across the Internet, identified by the prefix 2000::/3.
Its purpose is for link-local communication, identified by the prefix fe80::/10.
The correct answer is:

✅ **Its purpose is for global communication across the Internet, identified by the prefix 2000::/3.**

Why?

A Global unicast address (GUA) is:

- Publicly routable
- Used for communication across the Internet
- Similar to a public IPv4 address

Global Unicast Addresses typically use the prefix:

```
2000::/3
```

Why not the others?

- `fc00::/7` → Unique Local Address (ULA)
- `ff00::/8` → Multicast addresses
- `fe80::/10` → Link-local addresses

✅ **Final Answer: Its purpose is for global communication across the Internet, identified by the prefix 2000::/3.**

34. Question

Which security feature available on modern network switches allows the administrator to restrict the number of instances of a particular MAC address that can be seen on a given switch port?
Port Security
Port Mirroring
Virtual LANs (VLANs)
Access Control Lists (ACLs

The correct answer is:

✅ **Port Security**

Why?

Cisco **Port Security** allows administrators to:

- Limit the number of MAC addresses learned on a switch port
- Permit only specific MAC addresses
- Prevent unauthorized devices from connecting

Example uses:

- Restricting a port to one device
- Shutting down a port on violation
- Protecting against MAC flooding attacks

Why not the others?

- **Port Mirroring** → Copies traffic for monitoring/analysis
- **VLANs** → Segment broadcast domains
- **ACLs** → Filter traffic based on IP/MAC/protocol rules

✅ **Final Answer: Port Security**

35. Question

Which DHCP Snooping component helps to prevent DHCP spoofing attacks by ensuring that only responses from authorized DHCP servers are allowed?
Trusted Ports
Option 82 Insertion
Rate Limiting
DHCP Snooping Binding Database

The correct answer is:

✅ **Trusted Ports**

Why?

In Cisco DHCP snooping, **trusted ports** are interfaces allowed to send:

- DHCP Offer
- DHCP ACK
- Other server responses

Only ports connected to legitimate DHCP servers should be configured as trusted.

All other ports remain untrusted, which blocks rogue DHCP server responses and prevents DHCP spoofing attacks.

Why not the others?

- **Option 82 Insertion** → Adds relay agent information

- **Rate Limiting** → Protects against DHCP starvation attacks
- **DHCP Snooping Binding Database** → Stores IP-to-MAC bindings

✅ **Final Answer: Trusted Ports**

36. Question

In a multi-VLAN network, what is the purpose of using a VTP (VLAN Trunking Protocol) mode of operation set to 'transparent' on a Cisco switch?
The switch will neither advertise nor synchronize VLANs, but it will forward VTP advertisements.
The switch will advertise its own VLAN configuration and synchronize with other switches.
The switch will only synchronize VLANs but not advertise its own configuration.
The switch will only advertise VLAN configuration but will not synchronize with other switches.

The correct answer is:

✅ **The switch will neither advertise nor synchronize VLANs, but it will forward VTP advertisements.**

Why?

In VLAN Trunking Protocol **transparent mode**:

- The switch does **not participate** in VTP synchronization
- It maintains its own local VLAN database
- It forwards received VTP advertisements to other switches

This mode is often used when administrators want manual VLAN control while still allowing VTP traffic to pass through the switch.

Why not the others?

- Advertising and synchronizing VLANs → Server mode behavior
- Synchronize only → Client mode behavior
- Advertise only → Not how transparent mode works

✅ **Final Answer: The switch will neither advertise nor synchronize VLANs, but it will forward VTP advertisements.**

37. Question

Which of the following IPv6 transition mechanisms allows IPv6-only hosts to communicate with IPv4-only hosts by using a translation method at the network boundary?
NAT64
Dual Stack

ISATAP
6to4 Tunneling

The correct answer is:

✅ **NAT64**

Why?

NAT64 allows:

- IPv6-only hosts to communicate with IPv4-only hosts
- Translation between IPv6 and IPv4 packets at the network boundary

It works similarly to NAT in IPv4 environments but translates between the two IP versions.

Why not the others?

- **Dual Stack** → Devices run both IPv4 and IPv6 simultaneously
- **ISATAP** → IPv6 tunneling over IPv4 internal networks
- **6to4 Tunneling** → Encapsulates IPv6 traffic across IPv4 networks

✅ **Final Answer: NAT64**

38. Question

You are configuring an IEEE 802.1X authentication environment. Which component is responsible for validating the credentials of the devices attempting to access the network?
Authenticator
Authentication Server
Supplicant
Access Point

The correct answer is:

✅ **Authentication Server**

Why?

In IEEE 802.1X authentication:

- **Supplicant** → Client requesting access
- **Authenticator** → Switch or access point controlling access
- **Authentication Server** → Validates credentials

The authentication server is usually a:

- RADIUS server
- Cisco ISE server
- Microsoft NPS server

It checks usernames, passwords, certificates, or other credentials before granting network access.

Why not the others?

- Authenticator only relays authentication messages
- Supplicant provides credentials but does not validate them
- Access Point may act as authenticator, not the credential validator

✅ **Final Answer: Authentication Server**

39. Question

You are tasked with configuring a static default route on a router using the next-hop IP address. Which of the following commands correctly accomplishes this task?
ip default-gateway 192.168.1.1
ip route 0.0.0.0 0.0.0.0 192.168.1.1
default-route 0.0.0.0 192.168.1.1
ip route 0.0.0.0 0.0.0.0 Serial0/0

The correct answer is:

✅

```
ip route 0.0.0.0 0.0.0.0 192.168.1.1
```

Why?

On a Cisco router, this command creates a **static default route** using a next-hop IP address.

Breakdown:

- `0.0.0.0 0.0.0.0` → Matches all unknown destinations
- `192.168.1.1` → Next-hop router

This route is often called the:

- Gateway of last resort

Why not the others?

- `ip default-gateway` → Used mainly on Layer 2 switches, not routing-enabled routers
- `default-route` → Invalid Cisco IOS syntax
- `ip route ... Serial0/0` → Uses an exit interface, not a next-hop IP address

✅ **Final Answer:**

```
ip route 0.0.0.0 0.0.0.0 192.168.1.1
```

40. Question

Which of the following best describes the purpose of the OUI (Organizationally Unique Identifier) in an Ethernet MAC address?
It specifies the data payload type in the Ethernet frame.
It uniquely identifies the manufacturer of the network interface card.
It identifies the network protocol being used.
It indicates the subnet of the network

The correct answer is:

✅ **It uniquely identifies the manufacturer of the network interface card.**

Why?

In an Ethernet Media Access Control address address, the **OUI (Organizationally Unique Identifier)**:

- Is the first 24 bits (first 3 bytes) of the MAC address
- Identifies the hardware manufacturer/vendor

Example:

```
00:1A:2B
```

could identify a specific vendor assigned by IEEE.

Why not the others?

- Payload type is identified by the Ethertype field
- Network protocol is not identified by the OUI
- MAC addresses do not indicate IP subnets

✅ **Final Answer: It uniquely identifies the manufacturer of the network interface card.**

41. Question

In the implementation of WPA3, which of the following security enhancements is designed to protect against brute-force attacks by using Password-Authenticated Key Exchange (PAKE)?
Temporal Key Integrity Protocol (TKIP)
Advanced Encryption Standard (AES)
Protected Management Frames (PMF)
Simultaneous Authentication of Equals (SAE)

The correct answer is:

✅ **Simultaneous Authentication of Equals (SAE)**

Why?

Wi-Fi Protected Access 3 uses **SAE (Simultaneous Authentication of Equals)** as a major WPA3 enhancement.

SAE:

- Is based on a Password-Authenticated Key Exchange (PAKE)
- Protects against offline brute-force password attacks
- Replaces the WPA2 Pre-Shared Key (PSK) handshake

It is also known as the:

- Dragonfly Handshake

Why not the others?

- **TKIP** → Older WPA security mechanism, now outdated
- **AES** → Encryption algorithm, not PAKE authentication
- **PMF** → Protects management frames from spoofing/deauthentication attacks

✅ **Final Answer: Simultaneous Authentication of Equals (SAE)**

42. Question

You have been tasked with setting up a highly available DHCP service on a network that uses both IPv4 and IPv6 addressing. Given the requirements, which Cisco technology would be the best choice to ensure DHCP failover and load balancing, and how would it be implemented?
Implement DHCPv6 Relay and IPv4 DHCP Relay on multiple routers to distribute DHCP requests.
Use HSRP (Hot Standby Router Protocol) to provide a virtual IP for the DHCP server.
Implement DHCP Server Failover for IPv4 and configure DHCPv6 for stateless address auto-configuration with redundant servers.
Configure DHCPv4 and DHCPv6 on two separate routers and manually sync the lease databases

The correct answer is:

✅ **Implement DHCP Server Failover for IPv4 and configure DHCPv6 for stateless address auto-configuration with redundant servers.**

Why?

For high availability in mixed IPv4/IPv6 environments:

- DHCP failover for IPv4 provides:
    - Lease synchronization
    - Redundancy

    - Load balancing
    - Automatic failover
- For IPv6, networks commonly use:
    - Stateless Address Autoconfiguration (SLAAC)
    - Redundant DHCPv6 servers for additional services such as DNS information

This combination provides scalable and resilient address assignment.

Why not the others?

- DHCP relay only forwards requests and does not provide failover/load balancing
- HSRP protects gateway redundancy, not DHCP lease synchronization
- Manual synchronization is inefficient and unreliable

✅ **Final Answer: Implement DHCP Server Failover for IPv4 and configure DHCPv6 for stateless address auto-configuration with redundant servers.**

43. Question

Which command would you use to verify the path a packet takes through the network to a destination of 192.168.1.1 and what potential issues might you identify from its output?
show ip route 192.168.1.1
ping 192.168.1.1
show cdp neighbors
traceroute 192.168.1.1

The correct answer is:

✅ **traceroute 192.168.1.1**

Why?

On a Cisco device, the command:

```
traceroute 192.168.1.1
```

shows:

- The path packets take through the network
- Each router hop along the route
- Delay/latency information

It helps identify issues such as:

- Routing loops
- High latency

- Packet loss
- Unreachable hops
- Misconfigured routes

Why not the others?

- `show ip route` → Displays routing table only
- `ping` → Tests reachability, not the full path
- `show cdp neighbors` → Shows directly connected Cisco devices only

✅ **Final Answer: traceroute 192.168.1.1**

44. Question

Which of the following represents the CORRECT implementation of a MAC address table security feature to mitigate the threat of a CAM table overflow attack on a Cisco switch?
Using BPDU Guard on ports with PortFast enabled
Configuring Port Security with a maximum number of MAC addresses
Enabling IP Source Guard on all ports
Implementing Dynamic ARP Inspection (DAI)

The correct answer is:

✅ **Configuring Port Security with a maximum number of MAC addresses**

Why?

Cisco **Port Security** helps mitigate CAM table overflow attacks by:

- Limiting the number of MAC addresses learned on a switch port
- Preventing attackers from flooding the CAM table with fake MAC addresses
- Protecting against unauthorized devices

Example:

```
switchport port-security maximum 2
```

Why not the others?

- **BPDU Guard** → Protects against rogue switches/STP attacks
- **IP Source Guard** → Protects against IP spoofing
- **Dynamic ARP Inspection (DAI)** → Protects against ARP spoofing

✅ **Final Answer: Configuring Port Security with a maximum number of MAC addresses**

45. Question

Which AAA component can Cisco ISE use for centralizing network access policies and can enforce them using 802.1X, MAB, and WebAuth?
Auditing
Authorization
Authentication
Accounting

The correct answer is:

✅ **Authorization**

Why?

Cisco Cisco Identity Services Engine uses the AAA component **Authorization** to:

- Centralize network access policies
- Decide what users/devices are allowed to access
- Enforce policies through:
  - 802.1X
  - MAB (MAC Authentication Bypass)
  - WebAuth

Authorization determines:

- VLAN assignments
- ACLs
- Security group policies
- Access permissions

Why not the others?

- **Authentication** → Verifies identity only
- **Accounting** → Logs and tracks activity
- **Auditing** → General compliance/monitoring concept, not a core AAA function

✅ **Final Answer: Authorization**

46. Question

A network administrator is troubleshooting an issue where a PC cannot reach some devices on a different subnet, while other devices on the same subnet are reachable. The administrator notices that the router's ARP table has incomplete entries for some of the unreachable devices. Which of the following could be a possible reason for this behavior?
ARP inspection is misconfigured or not functioning correctly on the switch.
There is a VLAN misconfiguration on the switch connecting the subnets.

The devices have incorrect default gateways configured.
There is a routing loop between the subnets.

The correct answer is:

✅ **ARP inspection is misconfigured or not functioning correctly on the switch.**

Why?

Cisco incomplete ARP table entries usually indicate that:

- ARP requests are being sent
- But ARP replies are not being received or accepted

A misconfigured Dynamic ARP Inspection (DAI) setup can:

- Drop legitimate ARP replies
- Prevent ARP table completion
- Cause intermittent connectivity to some hosts

This would explain why:

- Some devices are reachable
- Others are not
- ARP entries remain incomplete

Why not the others?

- VLAN misconfiguration would usually affect broader connectivity
- Incorrect default gateways affect outbound routing, not ARP completion directly
- Routing loops cause instability, not specifically incomplete ARP entries

✅ **Final Answer: ARP inspection is misconfigured or not functioning correctly on the switch.**

47. Question

In a switched network, two switches, S1 and S2, are connected to each other on ports GigabitEthernet0/1 with the following configuration (reduced for brevity). Switch S1 runs Rapid PVST+ and has its root bridge priority set to 4096, while Switch S2 runs MST with its root bridge priority set to 0. Both switches have the same bridge ID. Which bridge will be the root bridge and why?
Neither switch will become the root bridge due to incompatibility between Rapid PVST+ and MST.
Switch S1 will be the root bridge because it has a lower root bridge priority.
Switch S2 will be the root bridge because it has a priority of 0, which is lower than 4096.

Switch S2 will be the root bridge because MST takes precedence over Rapid PVST+ in spanning tree calculations.

The correct answer is:

✅ **Switch S2 will be the root bridge because it has a priority of 0, which is lower than 4096.**

Why?

In STP/RSTP/MST, the root bridge is chosen by the **lowest Bridge ID**.

Bridge ID includes:

```
Bridge Priority + MAC Address
```

- S1 priority = **4096**
- S2 priority = **0**

Lower priority wins, so **S2 becomes root bridge**.

✅ **Final Answer: Switch S2 will be the root bridge because it has a priority of 0, which is lower than 4096.**

48. Question

In which scenarios would using the trunk allowed VLAN configuration command on a switch be most beneficial?
Enabling trunk mode on a switchport
Limiting VLAN traffic to specific VLANs on a trunk port
Assigning a native VLAN to a trunk port
Configuring a trunk port to support voice VLANs
The correct answer is:

✅ **Limiting VLAN traffic to specific VLANs on a trunk port**

Why?

On a Cisco switch, the command:

```
switchport trunk allowed vlan
```

is used to:

- Control which VLANs are permitted across a trunk link
- Reduce unnecessary broadcast traffic
- Improve security and traffic management

Example:

```
switchport trunk allowed vlan 10,20,30
```

Why not the others?

- Enabling trunk mode uses:

```
switchport mode trunk
```

- Native VLAN uses:

```
switchport trunk native vlan
```

- Voice VLAN uses:

```
switchport voice vlan
```

✅ **Final Answer: Limiting VLAN traffic to specific VLANs on a trunk port**

49. Question

In the context of deploying security measures on a Cisco network, which of the following features provided by Cisco Identity Services Engine (ISE) helps in dynamically enforcing security policies based on the user profile, device type, and posture assessment?
802.1X MAB
Port Security
Security Group Access (SGA)
TrustSec

The correct answer is:

✅ **TrustSec**

Why?

Cisco Cisco Identity Services Engine integrates with Cisco TrustSec to:

- Dynamically enforce security policies
- Use identity-based access control
- Consider:
  - User profile
  - Device type
  - Posture assessment
  - Location and role

TrustSec uses Security Group Tags (SGTs) for scalable policy enforcement across the network.

Why not the others?

- **802.1X MAB** → Authentication methods only

- **Port Security** → MAC-based switch port protection
- **Security Group Access (SGA)** → Related concept within TrustSec, but TrustSec is the broader Cisco solution

✅ **Final Answer: TrustSec**

50. Question

Which of the following methods is used to provide integrity and authentication for routing updates in OSPF?
IPsec
Public Key Infrastructure (PKI)
MD5 authentication
SHA-1 authentication
The correct answer is:

✅ **MD5 authentication**

Why?

Open Shortest Path First commonly uses **MD5 authentication** to:

- Verify the authenticity of OSPF routing updates
- Ensure message integrity
- Prevent unauthorized routers from injecting false routes

Routers sharing the same MD5 key can form trusted OSPF neighbor relationships.

Why not the others?

- **IPsec** → Can secure traffic but is not the standard OSPF update authentication method
- **PKI** → Certificate infrastructure, not typical OSPF authentication
- **SHA-1** → Used in some newer security contexts, but classic OSPF authentication commonly references MD5

✅ **Final Answer: MD5 authentication**

51. Question

In the context of HSRP (Hot Standby Router Protocol), what does the 'preempt' command achieve when configured on a router?
It resets the HSRP timers to their default values.
It stops the current active router from participating in the HSRP election.
It forces the router to wait before taking any active role in HSRP, regardless of its priority.
It allows the router to immediately take over as the active router if it has a higher priority.
The correct answer is:

✅ **It allows the router to immediately take over as the active router if it has a higher priority.**

Why?

Hot Standby Router Protocol `preempt` enables a router with a higher HSRP priority to reclaim the Active role when it becomes available again.

Without `preempt`:

- The current Active router remains Active even if a higher-priority router returns online.

With `preempt`:

- Higher-priority router automatically takes over as Active.

Example:

```
standby 1 preempt
```

Why not the others?

- It does not reset timers
- It does not stop participation in elections
- It does not delay active participation

✅ **Final Answer: It allows the router to immediately take over as the active router if it has a higher priority.**

52. Question

In a scenario where you have multiple VLANs across various switches, which of the following protocols should be used to efficiently manage VLAN information and dynamically configure VLANs on the switches?
RSTP
VTP
STP
LACP
The correct answer is:

✅ **VTP**

Why?

VLAN Trunking Protocol (VTP) is used to:

- Manage VLAN information centrally
- Dynamically distribute VLAN configurations across switches

- Simplify VLAN administration in large switched networks

VTP works over trunk links and helps reduce manual VLAN configuration on each switch.

Why not the others?

- **RSTP** → Prevents Layer 2 loops
- **STP** → Original spanning-tree protocol
- **LACP** → Aggregates multiple physical links into one logical link

✅ **Final Answer: VTP**

53. Question

Which of the following IPv6 address types is used to identify a set of interfaces, typically on different nodes, in such a way that packets sent to this address are delivered to only one of the interfaces, typically the closest one, according to the routing protocol's decision?
Anycast
Broadcast
Multicast
Unicast

The correct answer is:

✅ **Anycast**

Why?

Anycast addresses:

- Identify multiple interfaces on different devices
- Deliver packets to only one destination
- Usually the nearest or best destination based on routing metrics

Common uses include:

- DNS root servers
- Load distribution
- High availability services

Why not the others?

- **Broadcast** → Not used in IPv6
- **Multicast** → Sends packets to all members of a group
- **Unicast** → One-to-one communication

✅ **Final Answer: Anycast**

54. Question

Which of the following methods provides the most secure way to implement user authentication in a Cisco network environment?
Using local database authentication with strong passwords
Using CHAP for authentication on PPP links
Using RADIUS with encryption for both authentication and accounting
Using TACACS+ to encrypt the entire authentication process

The correct answer is:

✅ **Using TACACS+ to encrypt the entire authentication process**

Why?

Terminal Access Controller Access-Control System Plus (TACACS+) is considered highly secure because:

- It encrypts the entire authentication exchange
- Separates authentication, authorization, and accounting (AAA)
- Is commonly used for secure administrative access to Cisco devices

Why not the others?

- Local authentication is less centralized and scalable
- CHAP improves PPP security but is limited in scope
- RADIUS encrypts only the password portion, not the entire packet

✅ **Final Answer: Using TACACS+ to encrypt the entire authentication process**

55. Question

Which of the following is a primary function of the OSPF (Open Shortest Path First) designated router (DR) in a multi-access network?
To enable inter-VLAN routing in a Layer 3 switch environment
To generate and propagate the Link State Advertisements (LSAs) to and from other routers in the network
To act as a gateway between different OSPF areas
To block certain traffic based on ACLs (Access Control Lists)

The correct answer is:

✅ **To generate and propagate the Link State Advertisements (LSAs) to and from other routers in the network**

Why?

In Open Shortest Path First multi-access networks (such as Ethernet), the **Designated Router (DR)**:

- Reduces excessive OSPF neighbor relationships
- Collects and distributes LSAs
- Acts as the central exchange point for routing updates on the segment

This improves OSPF scalability and efficiency.

Why not the others?

- Inter-VLAN routing is unrelated to DR functionality
- Gateways between OSPF areas are ABRs (Area Border Routers)
- ACLs filter traffic and are unrelated to DR operations

✅ **Final Answer: To generate and propagate the Link State Advertisements (LSAs) to and from other routers in the network**

56. Question

You are configuring NAT on a router with the following requirements: translate internal addresses from 192.168.1.0/24 to a single public IP address, and allow multiple internal devices to share the single public IP address simultaneously. What type of NAT configuration should you use?
PAT (Port Address Translation)
Double NAT
Static NAT
Dynamic NAT

The correct answer is:

✅ **PAT (Port Address Translation)**

Why?

Port Address Translation allows:

- Multiple internal devices
- To share one public IP address
- By using different TCP/UDP port numbers

This is also called:

- NAT overload

Example scenario:

```
192.168.1.10 -> PublicIP:1025
192.168.1.11 -> PublicIP:1026
```

Why not the others?

- **Static NAT** → One-to-one mapping
- **Dynamic NAT** → Uses a pool of public IPs, not one shared IP
- **Double NAT** → Two layers of NAT, usually undesirable

✅ **Final Answer: PAT (Port Address Translation)**

57. Question

A network administrator has configured a router with the following Static NAT entry: ip nat inside source static 192.168.1.10 203.0.113.5. However, users are reporting that they cannot access the server located at 192.168.1.10 from the internet. What is a potential reason for this issue?
The access control list (ACL) is not permitting traffic to the server.
NAT is not enabled on the interface facing the internal network.
The router does not have an active route to the 192.168.1.0/24 network.
The public IP address is not properly configured on the router's WAN interface

The correct answer is:

✅ **NAT is not enabled on the interface facing the internal network.**

Why?

For Network address translation Static NAT to function properly, the router interfaces must be configured with:

- `ip nat inside`
- `ip nat outside`

If the internal interface is not marked as:

```
ip nat inside
```

the translation will not work correctly, even if the static NAT entry exists.

Example

```
interface GigabitEthernet0/0
 ip nat inside

interface GigabitEthernet0/1
 ip nat outside
```

Why not the others?

- ACLs could block traffic, but the question focuses on NAT configuration
- A route to 192.168.1.0/24 is usually directly connected internally
- The public IP in static NAT does not necessarily need to be configured directly on the WAN interface in all scenarios

✅ **Final Answer: NAT is not enabled on the interface facing the internal network.**

58. Question

In a large enterprise network with multiple VLANs, what is the primary function of the VTP pruning mechanism, and which of the following best describes its impact on network traffic?
VTP pruning reduces the number of broadcast, multicast, and flooded unicast packets transmitted across trunk links by restricting VLAN traffic only to those trunk links that the traffic is necessary for.
VTP pruning increases the security of the network by ensuring that VLAN information is only propagated to those devices that are authorized to receive it.
VTP pruning helps in balancing network traffic load by distributing VLAN traffic evenly across all available trunk links within the VTP domain.
VTP pruning ensures that VLAN configuration revisions are kept to a minimum by only transmitting configuration changes to switches that require them.

The correct answer is:

✅ **VTP pruning reduces the number of broadcast, multicast, and flooded unicast packets transmitted across trunk links by restricting VLAN traffic only to those trunk links that the traffic is necessary for.**

Why?

VLAN Trunking Protocol pruning improves network efficiency by:

- Preventing unnecessary VLAN traffic from crossing trunk links
- Reducing:
  - Broadcast traffic
  - Multicast traffic
  - Unknown unicast flooding

Traffic is forwarded only on trunk links where the VLAN is actually needed.

Benefits

- Conserves bandwidth
- Reduces unnecessary traffic

- Improves switched network performance

Why not the others?

- VTP pruning is not mainly a security feature
- It does not perform traffic load balancing
- It does not manage VTP revision numbers

✅ **Final Answer: VTP pruning reduces the number of broadcast, multicast, and flooded unicast packets transmitted across trunk links by restricting VLAN traffic only to those trunk links that the traffic is necessary for.**